

02-719 Week 1

Genomics and Epigenetics of the Brain

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Logistics

- Office hours: Wed. 1-2 PM, GHC 7711
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Motivation of Neurogenomics

Alzheimer's Disease

- More than 7 million Americans are currently living with Alzheimer's. (Projected to increase to 13 Million by 2050)
- 1 in 3 older adults dies with Alzheimer's or another dementia. It kills more than breast and prostate cancer combined.
- \$409 Billion dollars on Alzheimer's spendings in 2026, projected to be \$1 Trillion in 2050
- The lifetime risk for Alzheimer's at age 45 for women is 1 in 5, and for men it is 1 in 10. (Unknown reason)

Parkinson's Disease

- Lower prevalence than Alzheimer's (1 in 100 over age 60)
- 1M people in the United States, and 5M people worldwide affected

Addiction

- 46.3M people in the US have a substance use disorder in 2021.
- 105k drug overdose deaths per year

Major Depressive Disorder

- 13.8M - 15.4M adults affected
- One of the leading causes of suicide

Disease Categories

Most of these diseases fall into one of these following categories:

- No cure
- No effective long-term treatment (Alzheimer's, Parkinson's, other neurodegenerative diseases)
- Treatment comes with severe side effects (Bipolar, Major Depressive Disorder, ADHD, other psychiatric disorders)

Why are Brain Disorders So Hard to Treat?

The primary issue: What do you target when you treat a brain disorder?

- Target a specific brain region
- Specific cell types
- Specific hormones and neurotransmitters
- Specific proteins
- Specific section of DNA (gene therapy)
- Specific neural pathway

The problem is that modern medicines for brain diseases are like sledge hammers. One protein might influence the disease, but also regulate many other brain functions. Same goes with neurotransmitters, genes, etc. There isn't a single gene, protein, brain region, or biochemical pathway that equates to a single behavior.

Cellular Diversity of the Brain

The cellular diversity of the brain is overwhelming.

- Purkinje
- Microglia
- Astrocytes
- Bipolar
- Oligodendrocytes
- Schwann

Brief History of Neuroscience

- Santiago Ramon y Cajal (1852 - 1934): Traced neuron morphology and noticed there were different sub-types. Nobel prize winner.
- Cajal had a modern Zeiss microscope, which scientific discovery would not be possible without.
- Cajal's work was more "data-driven" than "hypothesis-driven". Rather than designing an experiment to test a specific theory of the brain, Cajal sought to characterize the diversity of neural cell types
- His work led to the neuron doctrine: "The brain is made of many different types of neurons."

Distinguishing Subtypes of Neurons

- Shape
- Firing
- Neurotransmitter
- Marker Genes
- Genomics

Single Neuron RNA-Seq measures genome-wide gene expression in single cells.